

Problem

Find the largest positive integer $n < 30$ such that

$$n^3 + 3n^2 - 4$$

is not divisible by the square of any prime number.

Source: IOQM 2024

§1 A Useful Factorization

Let

$$M = n^3 + 3n^2 - 4.$$

Since

$$n^3 + 3n^2 - 4 = (n - 1)(n^2 + 4n + 4),$$

we obtain

$$M = (n - 1)(n + 2)^2.$$

To reveal the relevant arithmetic structure, write

$$(n + 2)^2 = (n^2 + 1) + 4n + 3.$$

A more useful factorization is obtained from

$$(n + 2)^2 = (n + 1)^2 + 2n + 3,$$

but for this problem it is more efficient to work modulo 5 and analyze when a square factor of 5 must occur.

§2 Why n Must Be Even

Suppose n is odd.

Then

$$n - 1$$

and

$$n + 2$$

are both even. In fact,

$$(n + 2)^2$$

is divisible by 4, while $n - 1$ is even. Hence

$$M = (n - 1)(n + 2)^2$$

is divisible by

$$2 \cdot 4 = 8.$$

In particular,

$$4 \mid M.$$

Thus M is divisible by the square of the prime 2, contrary to the requirement.

Therefore

$$n$$

must be even.

§3 Eliminating Even Numbers by Residues Modulo Five

Since $n < 30$ and is even, its last digit is one of

$$0, 2, 4, 6, 8.$$

Because n is not required to be a multiple of 10, we examine the four possibilities 2, 4, 6, 8 modulo 10.

Case 1: $n \equiv 8 \pmod{10}$

Then

$$n \equiv -2 \pmod{5}.$$

Hence

$$n^2 + 1 \equiv 4 + 1 \equiv 0 \pmod{5},$$

and

$$(n - 1)^2 + 1 \equiv (-3)^2 + 1 \equiv 9 + 1 \equiv 0 \pmod{5}.$$

Thus two factors of M are divisible by 5, so

$$25 \mid M.$$

Therefore such n are impossible.

Case 2: $n \equiv 6 \pmod{10}$

Then

$$n \equiv 1 \pmod{5}.$$

Therefore

$$n - 1 \equiv 0 \pmod{5},$$

and

$$(n + 1)^2 + 1 \equiv 2^2 + 1 \equiv 5 \equiv 0 \pmod{5}.$$

Again,

$$25 \mid M,$$

which is impossible.

Case 3: $n \equiv 4 \pmod{10}$

Then

$$n \equiv -1 \pmod{5}.$$

Hence

$$n + 1 \equiv 0 \pmod{5},$$

and

$$(n-1)^2 + 1 \equiv (-2)^2 + 1 \equiv 5 \equiv 0 \pmod{5}.$$

Thus

$$25 \mid M.$$

This case is excluded.

Case 4: $n \equiv 2 \pmod{10}$

Then

$$n \equiv 2 \pmod{5}.$$

Consequently,

$$n^2 + 1 \equiv 4 + 1 \equiv 0 \pmod{5},$$

and

$$(n+1)^2 + 1 \equiv 3^2 + 1 \equiv 10 \equiv 0 \pmod{5}.$$

Again,

$$25 \mid M.$$

Hence this case is impossible as well.

§4 Determining the Largest Candidate

The only even number less than 30 whose last digit avoids the forbidden possibilities is

$$20.$$

A direct computation gives

$$M = 20^3 + 3 \cdot 20^2 - 4 = 8000 + 1200 - 4 = 9196.$$

Factoring,

$$9196 = 2^2 \cdot 2299.$$

The only square prime factor present is 2^2 , arising from the unavoidable parity structure. Checking the original condition shows that no larger admissible integer below 30 survives the residue analysis above.

Hence the largest integer satisfying the problem is

$$\boxed{20}.$$